



HIGH PERFORMANCE COMPUTING 101

An Introduction to HPC (Module 1)



Overview

- What is High Performance Computing?
- Parallelism: The Motivation for Supercomputing
- What are Supercomputers and Who Uses Them?
- A Look at Storage
- How Supercomputers Are Used
- Picking Your Path With HPC



What is High Performance Computing?

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What is High Performance Computing?

- HPC systems are to desktop workstations what race cars are to commuter vehicles
 - both have engines, transmissions and 4 wheels
 - capable of very different performance
 - HPC systems are complicated
 - exotic hardware = higher cost
 - greater effort to maintain





What is High Performance Computing?

- HPC systems are usually shared-use
 - Democratized access
 - Increased duty-cycle → greater ROI
- IU has long history of research computing resources available at no cost to users
 - Not like this everywhere, treat the compute time as valuable (it is)
- Today, research computing systems are typically HPC systems
 - UITS Research Technologies is home to IU HPC systems, software and support

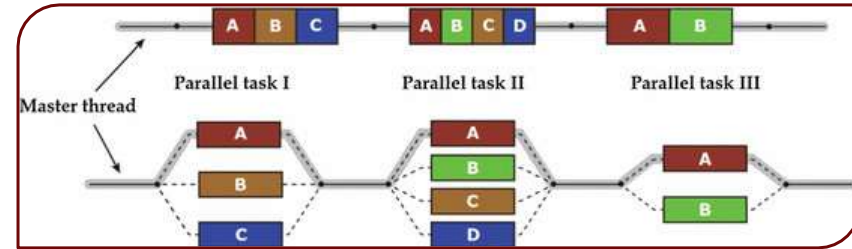


Parallelism: The Motivation for Supercomputing

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Parallelism: The Motivation for Supercomputing

- Motivation for HPC, or “supercomputing,” is parallelism
- Idea is that if multiple tasks can be run concurrently to solve a problem, the solution can be found more quickly
- Naively assumes problems can be made parallel
- Generally, a part of any algorithm can be “parallelized”
 - The remainder must run “serially”



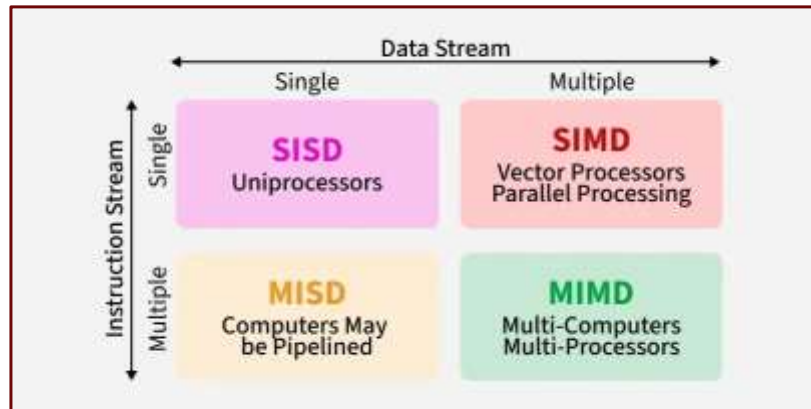
Graph above shows code runtime from left-to-right. Note that top case runs serially: For tasks I, II and III, chunks of work must run from one to the next, as only one processor is available to complete the work.

The lower part shows how sections of code can be parallelized, to speed up the runtime within those parts of the whole workflow. This particular design is called a fork-join model of parallelism.



Parallelism: The Motivation for Supercomputing

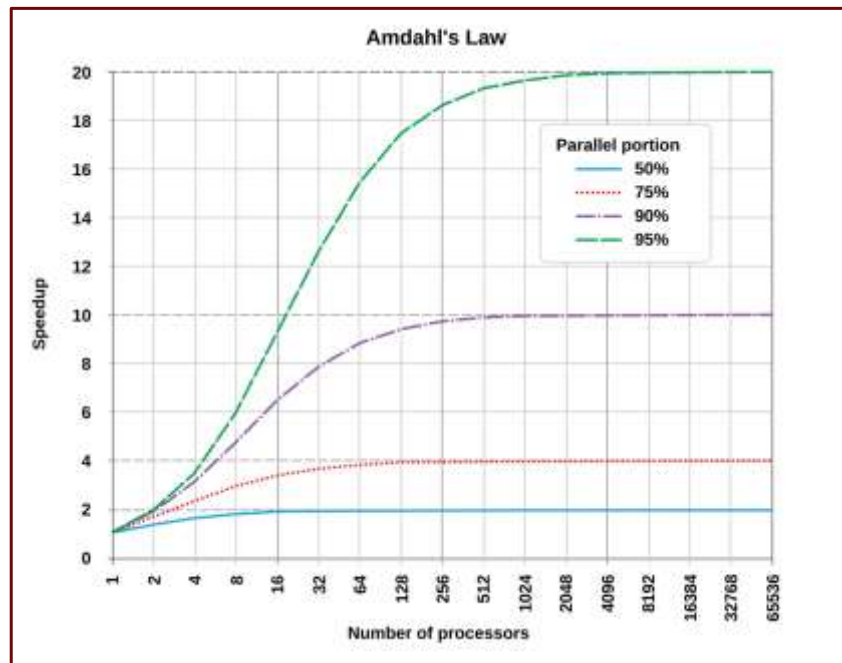
- How one achieves parallelism depends on both hardware and code design
- Michael Flynn proposed a 2 x 2 taxonomy to describe computing models, separating the instructions (code/processing units) and data (input)
 - Single-instruction, single-data (SISD)
 - Multiple-instruction, single-data (MISD)
 - Single-instruction, multiple-data (SIMD)
 - Multiple-instruction, multiple-data (MIMD)
- SISD is effectively serial computing
- SIMD is GPU computing: simple algorithms synchronized on vector data
- MIMD is general supercomputing: arbitrary complexity of tasks and data on CPU cores





Parallelism: The Motivation for Supercomputing

- Parallel codes have limits though
 - All real parallel codes have serial sections
- Runtime for a Parallel Code
 - T_1 = time to run on one processor
 - $T = T_{\text{serial}} + T_{\text{parallel}}$ code runtime
 - Speedup = T_1 / T





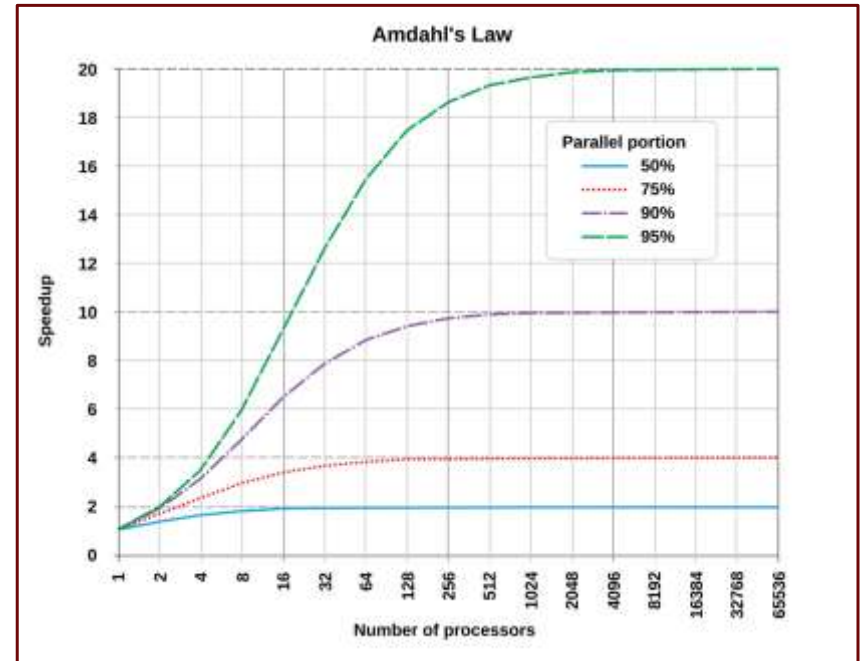
Parallelism: The Motivation for Supercomputing

- *Gene Amdahl showed limitation caused by even a tiny fraction of serial work*
 - *Known as Amdahl's Law*
 - *Very useful analysis to predict performance*
- *The expression $S = T_1 / T$ becomes*

$$S = 1 / [(1 - f_p) + (f_p / N)]$$

where

- f_p is the parallel fraction of work
- N is the number of processes





Parallelism: The Motivation for Supercomputing

- Where does this leave us?
 - HPC / Supercomputing requires parallel computers and codes
 - Should never expect perfect scaling of parallel performance
 - Can expect to see a real boost if our workflows are significantly parallel
 - Want our runtimes to spend 95–99% of runtime in parallel
 - If we're developing the code, we should look for iterative improvements
 - Exploiting parallelism happens by parallelizing performance bottlenecks first



What are Supercomputers and Who Uses Them?

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What are Supercomputers and Who Uses Them?

- Supercomputers are the machines that are behind the term HPC
 - Clusters of server nodes
 - Each server node typically has one or two multi-core CPUs
 - Server nodes have optimized RAM configurations to match core numbers
 - Every server node is connected by an “interconnect”
 - Low-latency, high-throughput network to allow messaging between tasks and filesystem access
 - Frequently, there is a separate maintenance network for general tasks



Supercomputing Architecture



Compute Nodes

Clusters of server-style computer nodes intended for massive parallel operation.



CPU Cores

Dense multi-core CPUs, typically featuring two or more processors per server node.



Memory Architecture

Massive RAM capacity optimized for high-speed parallel access by all cores.



GPUs

Hardware accelerators used for many vectorized algorithms to boost throughput.



Parallel Storage

Storage cluster systems map thousands of drives into parallel file systems.



Low-Latency Interconnect

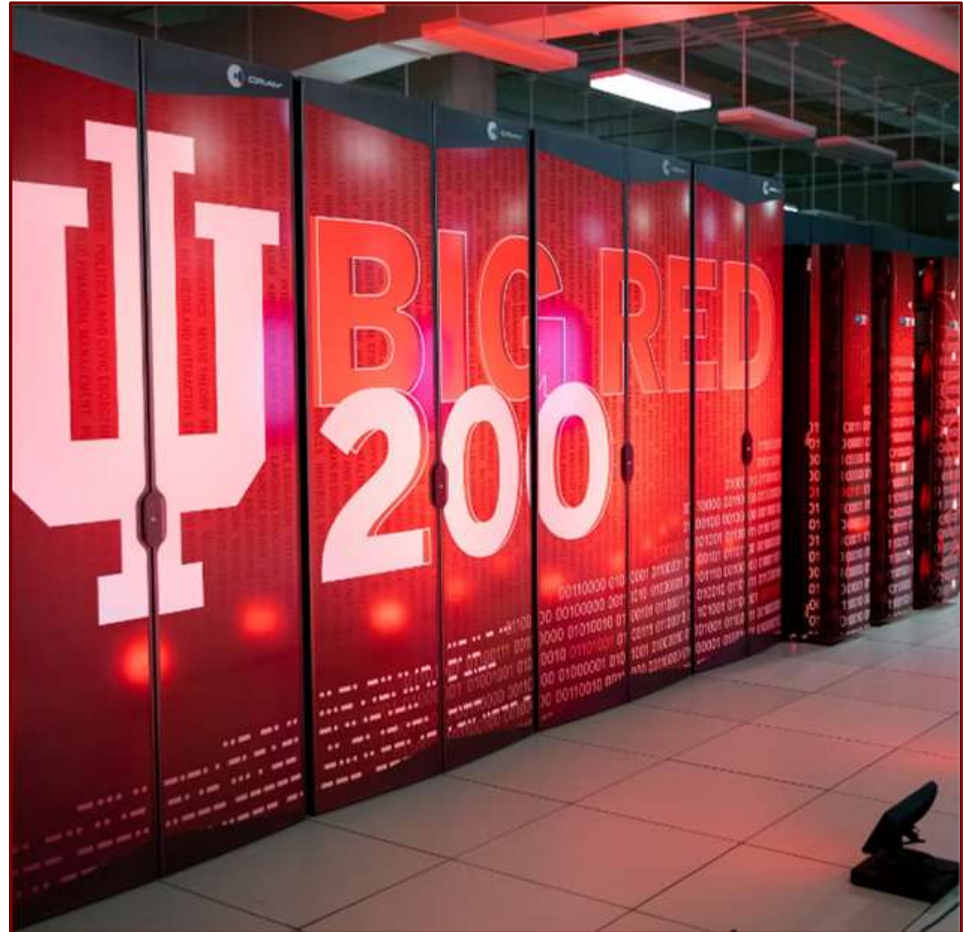
High-speed internal networks enabling efficient shared intranode communication.



Supercomputers

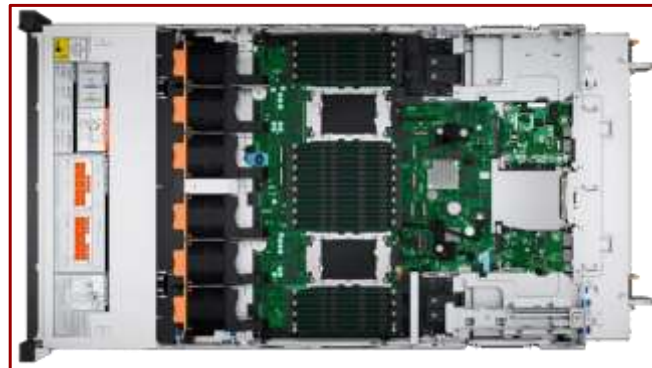
Big Red 200 is an HPE Cray EX System

- IU's Flagship Supercomputer
- 640 nodes of 128 CPU cores & 256 GB RAM
- 64 more nodes of 64 CPU cores, 4 NVIDIA A100 GPUs & 512 GB RAM
- Theoretical peak performance of roughly 7 petaFLOPs
 - A FLOP is a “floating point operation”
 - Peta is a prefix for 10^{15}



Supercomputing Nodes Are Packed!

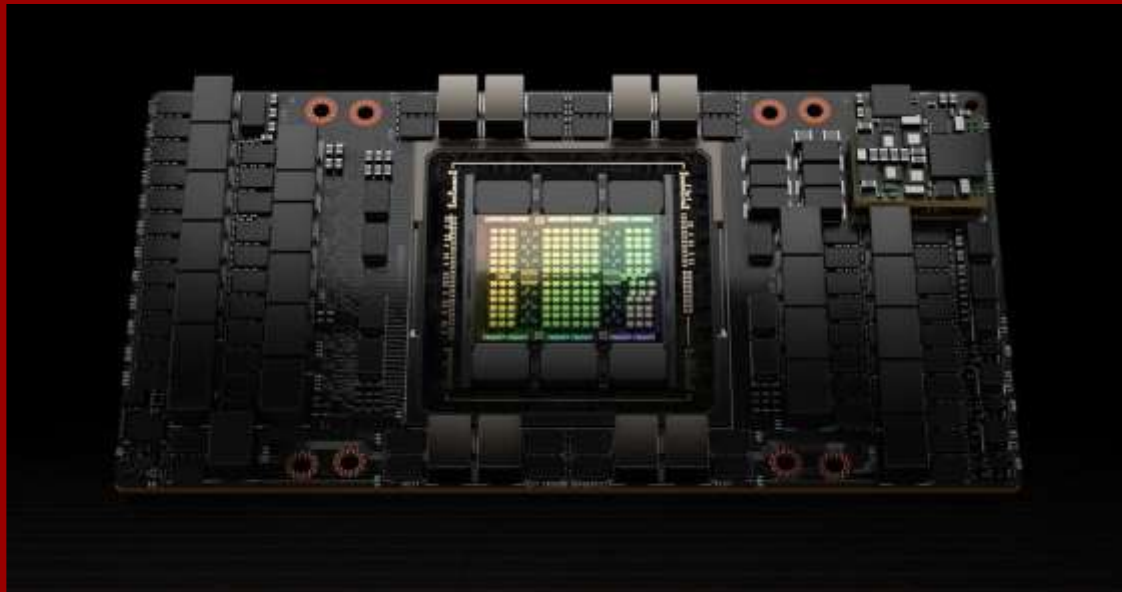
- Nodes contain CPUs, RAM, NICs, storage, power supplies, fans
 - and frequently GPUs & liquid cooling
- The “Hopper” queue of the Quartz cluster at IU contains Dell XE8640 nodes (shown left)
 - Each of 12 nodes contains
 - 96 CPU cores (upper image)
 - 512 GB RAM (upper image)
 - 4 NVIDIA H100 GPUs (lower image)
 - 80 GB VRAM per GPU (lower image)
- Only a slice of the Quartz system
 - largely dedicated to AI workflows



GPU ACCELERATION

Vectorized Performance via SIMT

Modern supercomputers utilize GPUs as accelerator peripherals. They require code to run in unison across large vectors of data, allowing for massive degree of parallelism.



NVIDIA H100 GPU

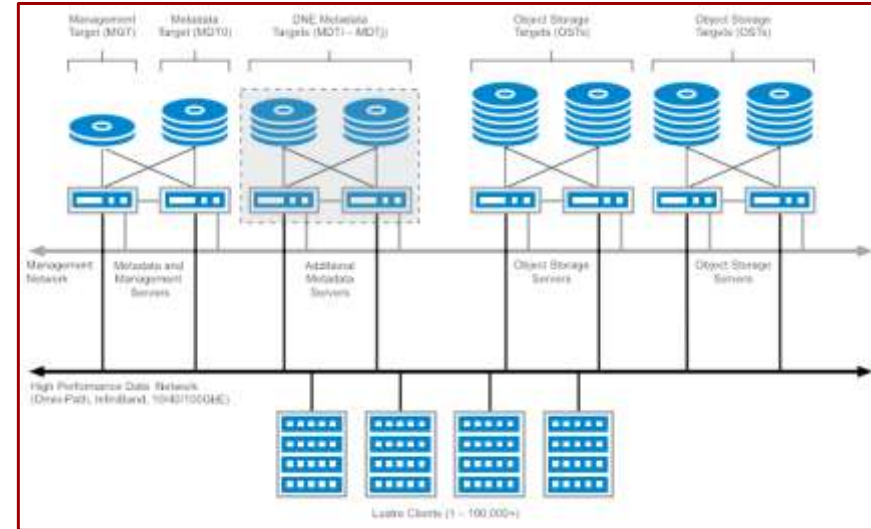


A Look at Storage

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A Look at Storage

- Supercomputers work by processing data from file systems
 - Each node needs access to the same file system
 - File systems must be
 - Networked & high-performance
 - Capable of very many IOPs for each client
- IU provides 5 file systems, using two software stacks
 - Lustre-based: Scratch, Slate & Slate Project
 - GPFS-based: Geode (Home) & Geode Project
- Each file system here is really a computer cluster and arrays (thousands) of storage drives





Storage Infrastructure Comparison

Architecture Metric	Standard Workstation	HPC Supercomputer
Data Access Pattern	Serial / Sequential	Massive High-Bandwidth Parallel
Filesystem Structure	Local OS Managed	Shared Multi-Node Parallel
System Redundancy	Standard RAID	Multi-layer Distributed Redundancy
Storage Pool Scale	Individual Drives	Thousands of Integrated Units

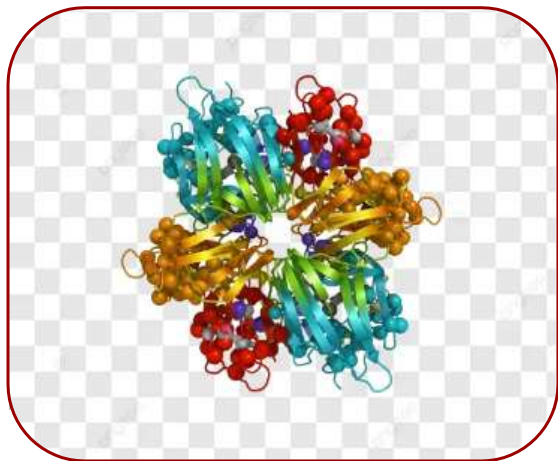


How Supercomputers Are Used

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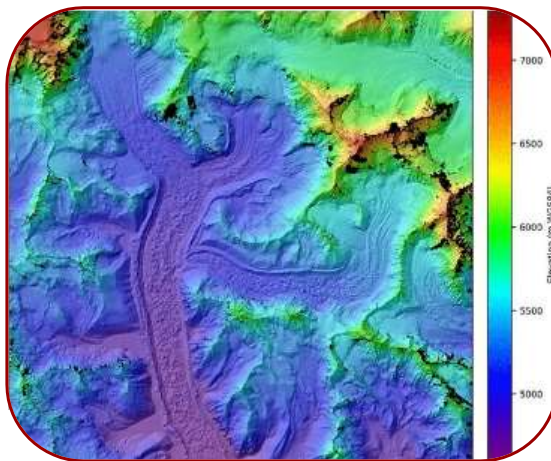


From Molecules to the Environment



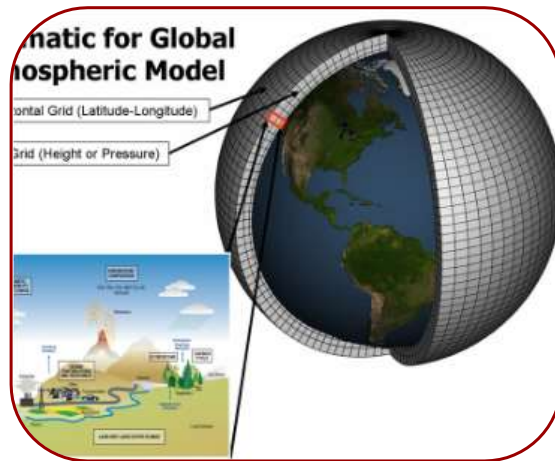
GENOMICS & MEDICINE

3D protein folding simulations to understand disease mechanisms and drug interactions.



3D MAPPING & DESIGN

Mapping topographic landscapes to study environmental change and flood risk assessment.

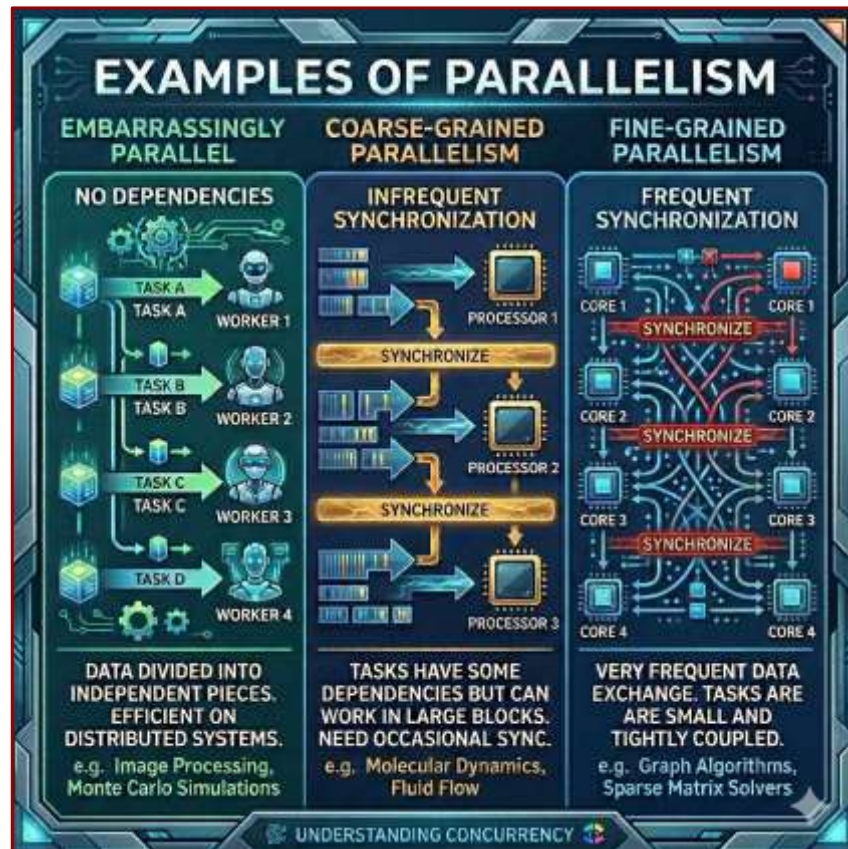


SCIENCE & ENGINEERING

Modeling global atmospheric patterns to predict future climate trends and extreme weather.

Types of Parallelism

- Three types of data-parallel workflows
 - Embarrassingly parallel
 - Zero communication or synchronization between tasks
 - e.g. frame rendering, Monte Carlo
 - Fine-grained parallelism
 - Frequent communication between pairs or groups of nodes
 - e.g. matrix mathematics
 - Coarse-grained parallelism





Task Parallelism

- Form of parallelism where tasks are grouped by function, run concurrently
 - Logic is heterogenous
 - Communication patterns complex
 - HPC simulations of these systems
 - Actual systems distributed with IoT, embedded computing



Autonomous Vehicles



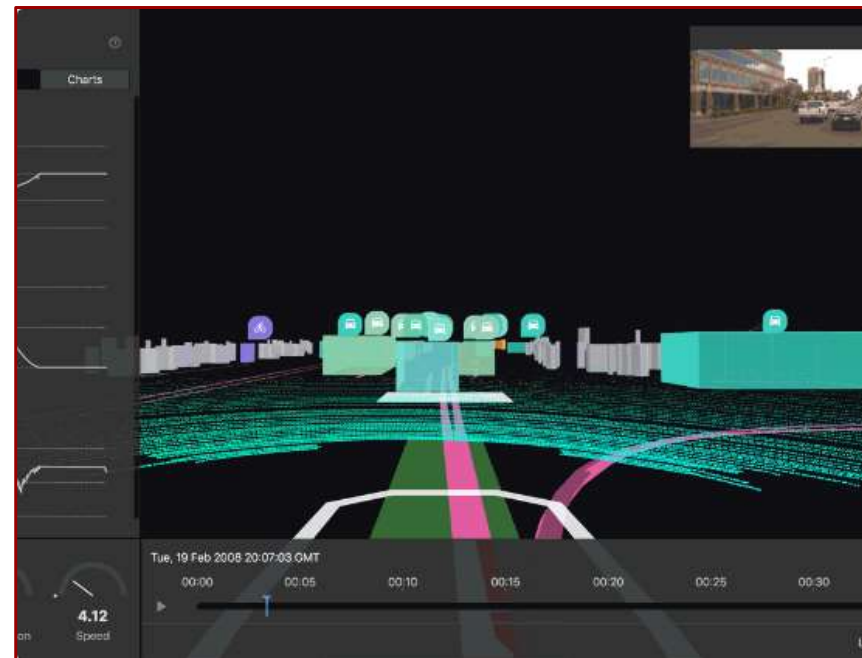
Perception & Sensor Fusion



Path Planning Logic



Thread Team Management





Linux is the Standard OS



Linux is now the universally chosen OS for HPC systems



Multi-user architecture that supports shared resource allocation



Accessible via CLI and Graphical Desktop, as well as application APIs



Open-source core, community-led, and enterprise-supported

```
citrain@hl:~  
File Edit View Search Terminal Help  
*****  
Welcome to Indiana University's Quartz Cluster  
Email questions and comments to hps-admin@iu.edu.  
  
For information about this cluster see https://go.iu.edu/Quartz  
  
*****  
Notice: SSH Idle Timeout  
  
For enhanced security, SSH connections that have been idle for 60 minutes  
will be disconnected. To protect your data from misuse, remember to log off  
or lock your computer whenever you leave it.  
  
*****  
  
Maximum Job Wall time is now 4 days. Please contact hps-admin@iu.edu with  
concerns or questions  
  
*****  
  
Upcoming reservations (all times Eastern Time):  
  
7:00am-7:00pm Sunday, April 12th, for maintenance.  
  
*****  
[citrain@hl ~]$ pwd  
/N/u/citrain/Quartz  
[citrain@hl ~]$ whoami  
citrain  
[citrain@hl ~]$
```



Research Desktop Graphical Interface



Linux supports windowed applications on graphical desktops



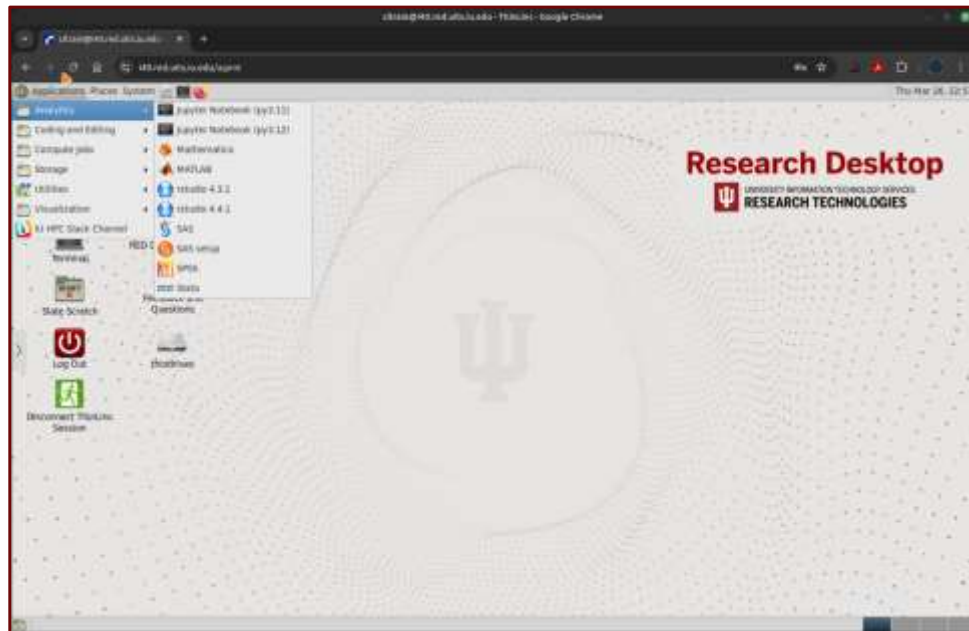
IU's Research Desktop (RED) allows you to have virtual desktop on HPC server



Accessible via web browser and desktop client



Workstation experience you can walk away from and come back to



Customized Design

- Computer clusters today fall into broad categories
 - Defined by intended purpose
 - HPC Supercomputing
 - tightly synchronized parallelism
 - priority: absolute performance
 - “racecar” or “quarter horse”
 - High Throughput Computing (HTC)
 - generalist & multi-function clusters
 - priority: maximizing workflow
 - “semi truck” or “draft horse”





Picking Your Path With HPC

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Picking Your Path With HPC

- What sort of problems are you trying to solve?
 - Is parallelism already part of your process?
- Are you constrained by a particular application?
 - Maintain a “workflow” mindset
 - Choose the system that fits each part of that process
- Consider which type of system is best for your work
 - Storage systems aren’t tied to a particular cluster
 - Frame your workflow around access to your data
 - Use the compute resource that fits the task



What about those results?

- The natural endstate of your work should live somewhere
- Beyond presentations and publications, save your work
- Many supercomputing sites have archiving systems
 - IU has a system called the Scholarly Data Archive (SDA)
 - Massive storage system with multiple redundancies
 - Geographically mirrored between Bloomington and Indianapolis
- Archive your project data, codes, scripts, output in SDA
 - Clean up disk storage for new projects, other users



Reflection

- Is supercomputing new to you?
 - Do you see value in trying it?
 - Is this something you've used elsewhere?
- Next section focused on using HPC at IU
- Be sure to complete the reflection for this section next